This 150-Million-Year-Old Bird Had Dinosaur Teeth and a Strange Secret

From: ScienceDaily Top Environment Mined: 2026-10-25 5 min read

A crow-sized **fossil** found in China is shaking up the story of how birds evolved. The 150-million-year-old creature, named Yuezhuornis yangyuanensis, lived in the Late Jurassic period, about 150 to 148 million years ago, and researchers say it may be the earliest known bird **lineage** to lose the ability to fly.

That is a surprise. Many people picture early birds steadily becoming better and better flyers. This **fossil** suggests something messier and more interesting: early birds were already trying out different ways of life, including life on the ground.

The **fossil** was found in the Zhenghe Fauna of Fujian Province in southeastern China and described in a study in Nature by researchers from the Institute of Vertebrate Paleontology and Paleoanthropology of the Chinese Academy of Sciences and the Fujian Institute of Geological Sciences.

Yuezhuornis mixes bird and dinosaur features in a striking way. It had bird-like pelvic and **hindlimb** structures, but its jaws held closely spaced, **serrated** teeth like those of **deinonychosaurs**, a group of small meat-eating dinosaurs related to birds. Its second toe could bend backward unusually far and carried a curved, sickle-shaped claw, another feature strongly linked to predatory dinosaurs such as dromaeosaurids and troodontids.

To place it on the family tree, scientists compared its anatomy using two methods, maximum parsimony and Bayesian tip-dating. Both put Yuezhuornis firmly within **Avialae**, the group that includes modern birds and their closest extinct relatives, just beyond Archaeopteryx and as a close relative of Zhengheornis.

Measurements of its limb proportions also pointed to the ground. They closely resembled those of modern ground-dwelling birds, suggesting Yuezhuornis lived on the ground and was likely unable to fly. Rather than a straight march toward flight, early bird evolution already included branches heading in the opposite direction.

Word Treasure

fossil -- The preserved remains of an ancient living thing.
lineage -- A line of living things descended from a shared ancestor.
serrated -- Having small, sharp teeth along an edge, like a saw.
Avialae -- The group that includes modern birds and their closest extinct relatives.
hindlimb -- A back leg.
deinonychosaurs -- A group of small meat-eating dinosaurs related to birds.

Background you need

Birds are living things descended from small meat-eating dinosaurs, so early birds often show a mix of bird and dinosaur features.

A fossil is the preserved remains of an ancient living thing, and it can show how bodies changed over time.

Scientists place a fossil on the family tree by comparing its bones with those of other living things.

Flight is not the only way of life for birds: some modern birds live mostly on the ground, and their leg shapes suit walking rather than flying.

Break it down

LEAD: A crow-sized fossil from China may be the earliest known bird lineage to lose flight

KEY EVENT: Researchers described Yuezhuornis yangyuanensis and placed it on the bird family tree

+- WHO: Yuezhuornis yangyuanensis, a 150-million-year-old creature from the Late Jurassic period

+- WHERE: The Zhenghe Fauna of Fujian Province in southeastern China

+- EVIDENCE: Bird-like pelvic and hindlimb structures, serrated teeth, and a sickle-shaped claw on the second toe

+- METHODS: Maximum parsimony and Bayesian tip-dating

CONCLUSION: Its limb proportions resembled modern ground-dwelling birds, suggesting life on the ground rather than a straight march toward flight

Why it matters

It shows bird evolution was not a straight march toward flight, but included branches trying different ways of life, which changes how scientists read the history of life on Earth.

Test what you remember

1. What is the name of the 150-million-year-old creature in this story?

- (A) Zhengheornis
- (B) Pseudorabdion
- (C) Archaeopteryx
- (D) Yuezhuornis yangyuanensis

2. Where was the fossil found?

- (A) Southeast Asia
- (B) The Zhenghe Fauna of Fujian Province in southeastern China
- (C) Northern China
- (D) Western Europe

3. What kind of teeth did Yuezhuornis have in its jaws?

- (A) A single large fang
- (B) No teeth at all
- (C) Flat grinding teeth
- (D) Closely spaced, serrated teeth like those of deinonychosaurs

4. What was special about its second toe?

- (A) It was missing
- (B) It was longer than its leg
- (C) It was webbed for swimming
- (D) It could bend backward unusually far and carried a curved, sickle-shaped claw

5. Where did both methods of comparison place Yuezhuornis?

- (A) Outside the bird family tree
- (B) Among modern flying birds only
- (C) Firmly within Avialae, just beyond Archaeopteryx
- (D) Among large meat-eating dinosaurs

6. What did measurements of its limb proportions suggest?

- (A) It lived on the ground and was likely unable to fly
- (B) It could swim long distances
- (C) It was a strong flyer
- (D) It lived in trees

Answer key (try the quiz first!): 1: D 2: B 3: D 4: D 5: C 6: A

Questions to think about

MY THOUGHTS (at least 20 words):